

Naoki Kimura

Ph.D. in Human-Computer Interaction nao-ki-mura.com | naoki.kimura.ac@gmail.com | [Google Scholar](#)

Selected Honors & Awards

- Microsoft Research Asia Fellowship (2020)
 - Google PhD Fellowship (2019)
 - ACM CHI Best Paper Honorable Mention (2019, 2018) – *Awarded to top 5% of submissions*
 - JST ACT-X Researcher (2019 – 2022) – *Grant: Frontier of Mathematics and Information Science*
 - Toyota / Dwango AI Scholarship (2018 – 2019)
 - JSPS Research Fellowship for Young Scientists (DC2) (2021)
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Education

The University of Tokyo | Tokyo, Japan *Ph.D. in Human-Computer Interaction* | 2019 – 2025 - Advisor: **Prof. Jun Rekimoto** - Graduate School of Interdisciplinary Information Studies

The University of Tokyo | Tokyo, Japan *M.S. in Applied Computer Science (Highest Distinction)* | 2013 – 2019 - Advisor: **Prof. Jun Rekimoto**

Professional Experience

LY Corporation Research (formerly Yahoo! JAPAN Research) | Tokyo, Japan *Chief Researcher* | Apr 2023 – Present - Leading advanced research projects in HCI and AI.

Google Research | Tokyo, Japan *Collaborative Researcher* | 2020 – 2021 - Advisor: **Dr. Shumin Zhai** - Research on mobile text entry and silent speech interfaces.

Microsoft Research Asia (MSRA) | Beijing, China (Remote) *Collaborative Researcher (PhD Fellow)* | 2020 - Advisor: **Dr. Frank Soong**

Georgia Institute of Technology | Atlanta, USA *Visiting Researcher* | 2019 – Present - Advisor: **Prof. Thad Starner** - Collaborating on wearable computing interfaces.

Preferred Networks (PFN) | Tokyo, Japan *Intern* | 2018 - Advisor: **Dr. Fabrice Matulic** - Contributed to the development of Chainer.

Selected Publications

[N. Kimura], “Rank-Residual Coding with a Shared Decoder: Lossless Low-Bit Communication for Ambiguous-KeyBoard AAC,” *IEEE SLT 2026*, 2026. **(Accepted)**

[N. Kimura], “3-KEY-INPUT: Exploring the Theoretical Minimum Keys for Text Entry,” *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2026)*, 2026.

[N. Kimura], T. Taniguchi, T. Starner, and J. Rekimoto, “HMDspeller: Non-visual Text Entry for Head-Mounted Displays,” *ACM CHI Conference on Human Factors in Computing Systems (CHI 2023)*, Hamburg, Germany, April 2023.

[N. Kimura], K. Higuchi, and J. Rekimoto, “Self-Supervised Learning for Silent Speech Interfaces,” *ACM Symposium on User Interface Software and Technology (UIST 2022)*, Bend, OR, USA, Oct 2022.

[N. Kimura], T. Nakahira, and J. Rekimoto, “SilentSpeller: Towards Silent Speech Interface with Ultra-Low-Power Silent Voice Recognition,” *ACM CHI Conference on Human Factors in Computing Systems (CHI 2022)*, New Orleans, LA, USA, April 2022.

[N. Kimura], S. Zhai, and J. Rekimoto, “Mobile SilentSpeller: Silent Speech Interface for Mobile Devices,” *ACM CHI Conference on Human Factors in Computing Systems (CHI 2021)*, Yokohama, Japan, May 2021.

[N. Kimura], Z. Zhao, and F. Soong, “End-to-End Silent Speech Recognition from EMG Signals,” *INTERSPEECH 2020*, Shanghai, China, Oct 2020.

[N. Kimura], M. Kono, and J. Rekimoto, “SottoVoce: An Ultrasound Imaging-Based Silent Speech Interaction Using Deep Neural Networks,” *ACM CHI Conference on Human Factors in Computing Systems (CHI 2019)*, Glasgow, UK, May 2019. **(Honorable Mention Award)**

[N. Kimura] and J. Rekimoto, “ExtVision: Augmentation of Visual Experiences with Generation of Context Images for Peripheral Vision Using Deep Neural Network,” *ACM CHI Conference on Human Factors in Computing Systems (CHI 2018)*, Montreal, Canada, April 2018. **(Honorable Mention Award)**

Skills

- **Languages:** Japanese (Native), English (Professional Proficiency)
- **Technical:** Python, PyTorch, Chainer, C++, Unity, Signal Processing (EMG, Ultrasound), LLM Integration